

Due on tuesday, February 7th, in class, or at the beginning of the first tutorial, whichever occurs earlier.

Problems marked **[SELF]** are for self-study and additional exercise; will not be marked.

----- ★ -----

The relativistic factor

$$\gamma(v) = \frac{1}{\sqrt{1 - (v/c)^2}}$$

is central in special relativity. It is a function of the speed (magnitude of velocity)  $v$ . It's vital to know the properties of  $\gamma(v)$  in some detail.

If tired of carrying along factors of  $c$ , you could use the common notation  $\beta = \frac{v}{c}$ . If there are multiple speeds involved, such as  $u$  and  $v$ , you could use  $\beta_u = \frac{u}{c}$  and  $\beta_v = \frac{v}{c}$ . But please express final results in terms of  $u$ ,  $v$  and  $c$  again.

----- ★ -----

1. (a) **[2 pts.]** Show that

$$\frac{1}{\gamma^2} + \frac{v^2}{c^2} = 1 \quad \left\{ \begin{array}{l} \text{Here } \gamma \text{ is used as} \\ \text{shorthand for } \gamma(v) \end{array} \right.$$

- (b) **[2 pts.]** Show that

$$\gamma(v) \times (c + v) = c \sqrt{\frac{c + v}{c - v}}$$

- (c) **[6 pts.]** Show that

$$\frac{d\gamma}{dt} = \frac{1}{c^2} \gamma^3 v \dot{v} \quad \text{and} \quad \frac{d\gamma}{dt} = \frac{1}{c^2} \gamma^3 \vec{v} \cdot \vec{a}$$

where we use the shorthands  $\gamma = \gamma(v)$  and  $\dot{v} = \frac{dv}{dt}$ . Note that  $\dot{v}$  is the derivative of the speed, not the derivative of the 3-vector  $\vec{v}$ , i.e.,  $\dot{v}$  is not the acceleration vector  $\vec{a} = \frac{d\vec{v}}{dt}$ .

- (d) **[5 pts.]** If  $w = (u + v)/(1 + uv/c^2)$ , derive a relation of the form

$$\gamma(w) = \gamma(u)\gamma(v) \times \text{something}$$

and state explicitly what this something is.

2. **[3 pts.]** Present a neat hand-drawn plot of  $\gamma(v)$  as a function of  $v$  or  $v/c$ . (As usual, you can use a computer plotting programme to figure out what this looks like, but please submit a hand-drawn plot.)

Looking at your curve, I should be able to immediately see the value of  $\gamma(v)$  at zero speed, the slope at zero speed, and at which speed, if any,  $\gamma(v)$  diverges.

3. The binomial series

$$(1+x)^\epsilon = 1 + \epsilon x + \frac{1}{2!}\epsilon(\epsilon-1)x^2 + \frac{1}{3!}\epsilon(\epsilon-1)(\epsilon-2)x^3 + \dots$$

holds for all  $|x| < 1$ . Please find a way to remember this, so that you can use it without having to look it up. (Look up ‘binomial series’ online.)

- (a) **[3 pts.]** Derive the binomial expansion using the Taylor expansion. State clearly which function you are expanding and around what point.
  - (b) **[2 pts.]** Expand  $(P+Q)^\epsilon$ , first assuming  $|Q| < |P|$ , and then for the case  $|P| < |Q|$ .
  - (c) **[3 pts.]** Use the binomial expansion to find good approximations for  $(1001)^{1/3}$  and  $(15)^{1/4}$ , without using a calculator.
  - (d) **[SELF]** For which values of  $\epsilon$  does the series terminate after a finite number of terms?
  - (e) **[4 pts.]** Expand  $\gamma$  in powers of  $v/c$ , including at least up to order  $(v/c)^6$ . This expansion is useful for small  $v$ , when  $\gamma$  is close to 1.
  - (f) **[3 pts.]** If you truncate the series, you get an approximant for  $\gamma(v)$ . Plot  $\gamma(v)$  as a function of  $v/c$ , together with the approximants obtained by keeping the first one, two, three or four terms of the series. This should show how successive approximants do a better and better job of approximating our  $\gamma(v)$  function.  
Please submit only a neat hand-drawn plot, no computer printouts.
  - (g) **[4 pts.]** Express  $v$  as a function of  $1/\gamma$ . Expand using the binomial expansion, at least up to order  $1/\gamma^6$ . For which values of  $v$  is this expansion useful?
4. **[3 pts.]** A low-flying earth satellite travels at about 8000 m/s. What is the factor  $\gamma$  for this speed?

5. We will be dealing a lot with *hyperbolic functions*.

(a) [3 pts.] Look up (or recall) and report the definitions of the hyperbolic functions  $\cosh(x)$ ,  $\sinh(x)$  and  $\tanh(x)$ , in terms of the exponential functions  $e^x$  and  $e^{-x}$ .

(b) [3 pts.] Provide neat hand-drawn plots of the three hyperbolic functions.

(You can use a computer program if you need to, but you really should be able to figure out how these functions look like, from the definitions in terms of  $e^x$  and  $e^{-x}$ . I suggest trying first without a computer program or textbook.)

(c) [4 pts.] Express the corresponding three trigonometric functions in terms of  $e^{ix}$  and  $e^{-ix}$ . Are the relations

$$\sinh(x) = c_1 \sin(ix) \quad \text{and} \quad \sinh(ix) = c_2 \sin(x)$$

correct for some value of  $c_1$  and  $c_2$ ? Which values?